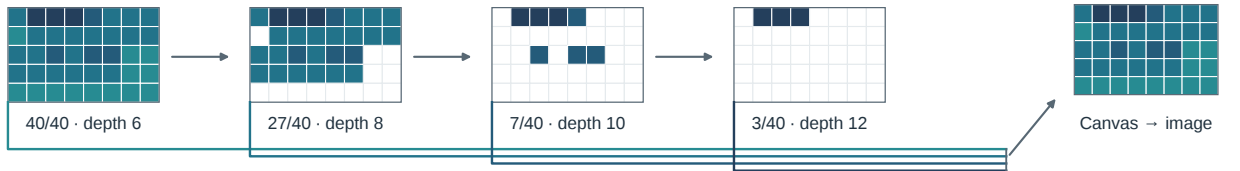


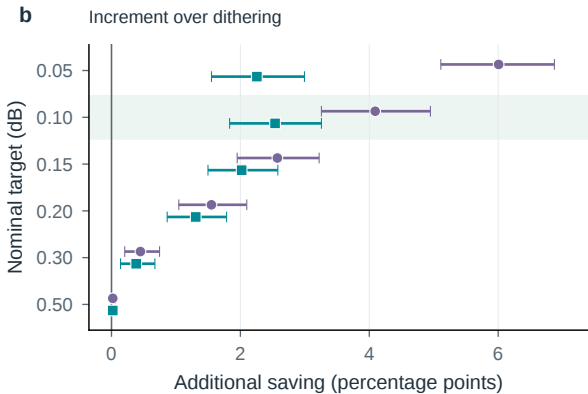
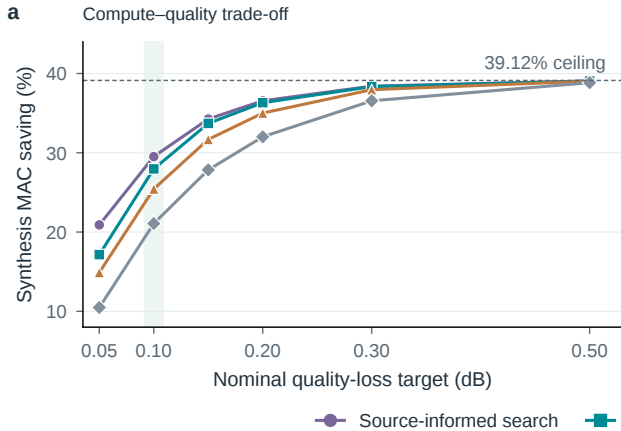
a Recover once; assign a stopping depth to each tile

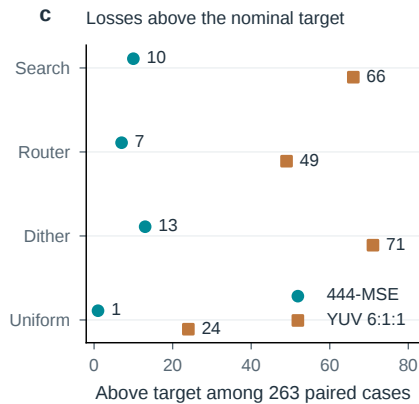
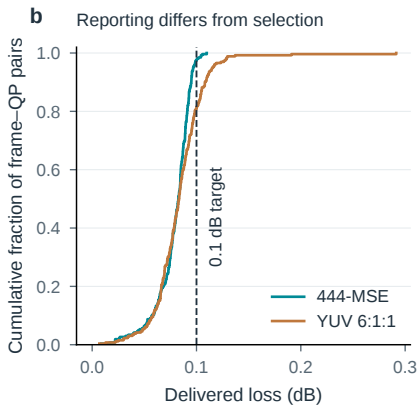
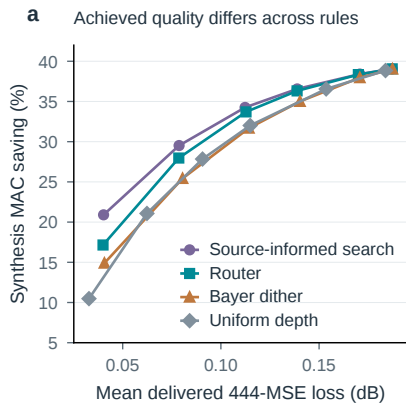


b Continue only the surviving tiles

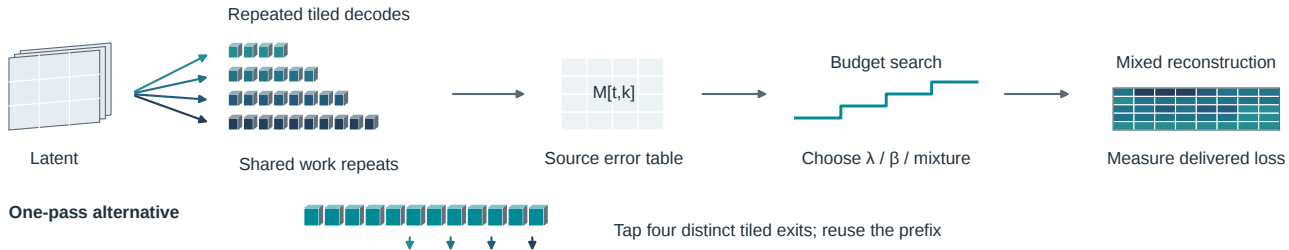


Pointwise exit adapters; deepest exit uses identity

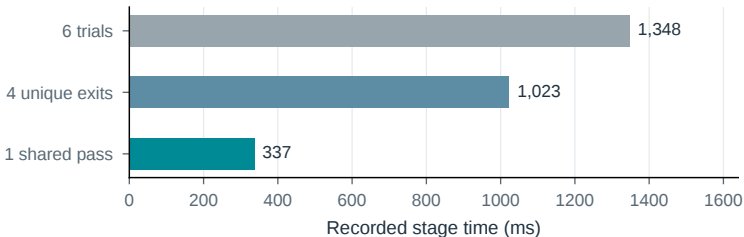




a The source is available at the encoder, but the error table must be computed



b Recorded table construction time



Historical timing: 8 frames $\times$ 3 repetitions at QP32. No new end-to-end latency measurement.

c Calibration changes the cost

Fixed validation-selected β

Router uses decoded features.
No per-frame source table is required.

Per-frame source-calibrated β

The plotted budget sweep uses $M[t,k]$.
A fast router does not remove that cost.

a

Source luma

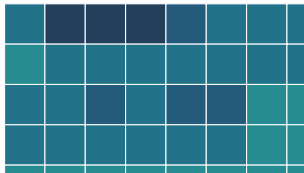


videoSRC05

QP32

b

Router

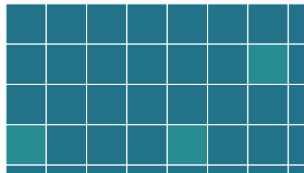


26.1% MAC saved

0.085 dB loss

c

Bayer dither

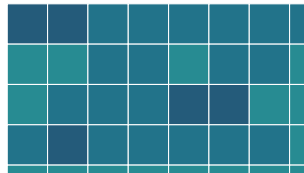


25.3% MAC saved

0.092 dB loss

d

Source-informed search



28.8% MAC saved

0.085 dB loss



6



8



10



12

executed trunk blocks

Only surviving tiles run the next block pair

Executed depth

Active spatial tiles

Exit

6

40

tiles



13

8



27

tiles



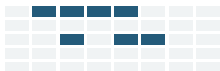
20

10



7

tiles



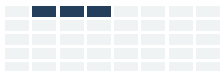
4

12



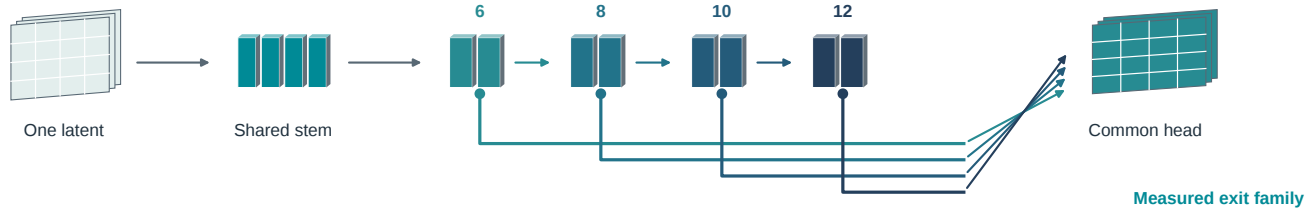
3

tiles



3

a Shared early exit: one representation, nested synthesis



b Independent depth controls: each codec learns its own representation



c Later routing study: select before encoding; signal which expert owns the payload



